Study notes: Abstract Algebra I, Module 2

Fraleigh, *A First Course in Abstract Algebra* (8th ed.), Module 2 (sections 5, 6 and 8 to 8.18). The syllabus names the 7th edition; items are matched by content. Numbers are the book's own. Student notes in my own words. Every definition, theorem, example, table and figure is kept with its book number. Exercises: the book's exercise sets are not in these notes, except the exercises the text names (Section 2 Exercise 20; Section 5 Exercises 51 and 57; Section 6 Exercise 54; Section 8 Exercise 54), each printed directly after the item that names it. Figures are described with their labels; they are not redrawn. Item 8.19 onward (alternating groups) is Module III and is not in these notes.

Section 5: Subgroups

Printed pp. 52–56 (5.1–5.26; Tables 5.8, 5.9; Figures 5.10, 5.27)

Before you start

- The four group properties: closure, associativity, identity and inverses (Section 2 of the book, Definition 2.1).
- Matrices and determinants: $\det(AB) = \det(A)\det(B)$, and a matrix is invertible exactly when its determinant is not 0 (Example 5.14 uses this).
- Complex numbers: modulus and the unit circle (Examples 5.5 and 5.26).

1. What a subgroup is

Subgroups inside groups (Definition 5.1)

Some groups sit inside bigger groups. The integers under addition sit inside the real numbers under addition. The positive rationals under multiplication do *not* sit inside the real numbers under addition, even though every positive rational is a real number. The reason is that the operation changes: $2 \cdot 3 = 6$ in the first group, but $2 + 3 = 5$ in the second. A subgroup must keep the same operation.

> **DEFINITION 5.1** If a subset H of a group G is closed under the operation of G, and H with that induced operation is itself a group, then H is a **subgroup of G**. Write $H \leq G$ for "H is a subgroup of G", and $H < G$ when $H \leq G$ and $H \neq G$.

In short: H is a subgroup when the group operation of G keeps you inside H, and H satisfies the group axioms by itself.

$\langle \mathbb{Z}, + \rangle < \langle \mathbb{R}, + \rangle$. But $\langle \mathbb{Q}^+, \cdot \rangle$ is not a subgroup of $\langle \mathbb{R}, + \rangle$, because the operation is different, even though $\mathbb{Q}^+ \subset \mathbb{R}$ as sets. Every group G has G itself and $\{e\}$ as subgroups.

> **DEFINITION 5.2** The subgroup G of G itself is the **improper subgroup**. All other subgroups are **proper**. The subgroup $\{e\}$ is the **trivial subgroup**. All other subgroups are **nontrivial**.

Examples of subgroups

Example 5.3. In the additive group $\mathbb{R}^n$ of n-component row vectors, the vectors whose first component is 0 form a subgroup.

Example 5.4. $\mathbb{Q}^+$ under multiplication is a proper subgroup of $\mathbb{R}^+$ under multiplication.

Example 5.5. The nth roots of unity U_n form a subgroup of U, the complex numbers of absolute value 1. And U is in turn a subgroup of $\mathbb{C}^*$, the nonzero complex numbers under multiplication.

Example 5.6. D_n is the set of one-to-one maps ϕ of $\mathbb{Z}_n$ onto itself that keep the edges of the regular n-gon P_n. D_n is a subset of $S_{\mathbb{Z}_n}$, and both are groups under composition, so $D_n \leq S_{\mathbb{Z}_n}$.

Example 5.7. There are two different groups of order 4 (Exercise 20 of Section 2, printed right after this example). The first is $\mathbb{Z}_4$. The second is the Klein 4-group V. Their tables are Tables 5.8 and 5.9.

EXERCISE 20, SECTION 2 (NAMED IN EXAMPLE 5.7) This exercise shows that there are two nonisomorphic group structures on a set of 4 elements. Let the set be $\{e, a, b, c\}$, with e the identity for the operation. A group table must begin as in Table 2.29 (Module 1 excerpt, printed page 31). The square with the question mark cannot be a. It must be e, or an element different from e and a; without loss of generality take that element to be b. If the square is e, the table can be completed in two ways to give a group. Find these two tables. (You need not check associativity.) If the square is b, the table can be completed in one way. Find that table. (Again, you need not check associativity.) Of the three tables, two give isomorphic groups. Say which two, and give the one-to-one onto relabelling that is an isomorphism.

a. Are all groups of 4 elements commutative?

b. Find a way to relabel the four matrices $\left\{\begin{bmatrix}1&0\\0&1\end{bmatrix}, \begin{bmatrix}0&-1\\1&0\end{bmatrix}, \begin{bmatrix}-1&0\\0&-1\end{bmatrix}, \begin{bmatrix}0&1\\-1&0\end{bmatrix}\right\}$ so that the matrix multiplication table is identical to one you constructed. This shows that the table defines an associative operation.

c. Show that for a particular value of n, the group elements of Exercise 14 can be relabelled so that their table is identical to one you constructed.

The only nontrivial proper subgroup of $\mathbb{Z}_4$ is $\{0, 2\}$. The set $\{0, 3\}$ is *not* a subgroup, since $3 + 3 = 2$ and $2 \notin \{0, 3\}$. The group V has three nontrivial proper subgroups: $\{e, a\}$, $\{e, b\}$ and $\{e, c\}$. The set $\{e, a, b\}$ is not a subgroup, because $ab = c$ and c is not in the set.

Table 5.8 (the group $\mathbb{Z}_4$ under $+$)

$+$	**0**	**1**	**2**	**3**
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table 5.9 (the Klein 4-group V)

V	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Subgroup diagrams

A *subgroup diagram* shows which subgroups sit inside which. A line going down from G to H means H is a subgroup of G. The bigger group goes higher.

Figure 5.10 shows two diagrams. In (a), $\mathbb{Z}_4$ is at the top, then $\{0, 2\}$, then $\{0\}$ at the bottom: a single chain. In (b), V is at the top, the three subgroups $\{e, a\}$, $\{e, b\}$, $\{e, c\}$ sit in the middle row, and $\{e\}$ is at the bottom.

Notice that if $H \leq G$ and $a \in H$, then the equation $ax = a$ has one solution in H, the identity of H. It also has one solution in G. These must be the same element, so the identity of H is the identity of G. The same kind of argument shows that the inverse of a in H is the inverse of a in G.

Checking whether a subset is a subgroup

Example 5.11. The continuous real-valued functions form a subgroup of the additive group of all real-valued functions. The sum of two continuous functions is continuous, the zero function is continuous, and the negative of a continuous function is continuous.

We want a routine that works for every example. The next theorem gives it.

THEOREM 5.12 A subset H of a group G is a subgroup of G if and only if:

1. H is closed under the operation of G;
2. the identity e of G is in H; and
3. for every $a \in H$, $a^{-1} \in H$.

Proof in brief. If H is a subgroup, the three conditions hold by definition. In the other direction, conditions 2 and 3 give the identity and inverse axioms in H. Associativity is inherited from G, because the equation $(ab)c = a(bc)$ is already true in G.

Example 5.13. The differentiable functions form a subgroup of the additive group of all real-valued functions, by the same argument as Example 5.11.

Example 5.14. In the multiplicative group G of invertible $n \times n$ complex matrices, the matrices of determinant 1 form a subgroup T. The reason: $\det(AB) = \det(A)\det(B)$ keeps T closed under multiplication, I_n has determinant 1, and if $\det(A) = 1$ then $\det(A^{-1}) = 1$. Theorem 5.12 then applies.

PLUS TWO RECALL: DETERMINANT OF A PRODUCT (STATUS OPEN, CHECK AGAINST THE KERALA TEXT)

Recall. For square matrices A and B of the same size, $\det(AB) = \det(A)\det(B)$. A square matrix A is invertible exactly when $\det(A) \neq 0$.

Re-derive.

1. Take $B = A^{-1}$. Then $\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det(I_n) = 1$.
2. So $\det(A) = 1$ forces $\det(A^{-1}) = 1$.

Saved. Example 5.14 (the matrices of determinant 1 form a subgroup).

Theorem 5.15 gives another test for a nonempty set. It says: a nonempty subset H of G is a subgroup if and only if ab^{-1} lies in the set whenever a and b do. The book states the theorem with G in both places, as printed; the condition that is meant is "$a, b \in H$ and $ab^{-1} \in H$". Its proof is Exercise 51. In practice Theorem 5.12 is about as quick.

EXERCISE 51, SECTION 5 (THE PROOF OF THEOREM 5.15) Prove Theorem 5.15.

Theorem 5.16 is the fastest test when the set is finite. A finite nonempty subset of G is a subgroup if and only if it is closed under the operation of G. Its proof is Exercise 57.

EXERCISE 57, SECTION 5 (THE PROOF OF THEOREM 5.16) Prove Theorem 5.16.

Example 5.17. U_n is a finite nonempty subset of $\mathbb{C}^*$, and if $z_1, z_2 \in U_n$ then $(z_1 z_2)^n = 1$. So U_n is closed, and by Theorem 5.16 it is a subgroup.

Example 5.18. The set $H = \{\iota, \rho, \rho^2, \ldots, \rho^{n-1}\}$ is a subgroup of D_n, because $\rho^k \rho^r = \rho^{k+nr}$ stays in H.

2. Cyclic subgroups

Building a subgroup from one element

Suppose a subgroup H of $\mathbb{Z}_{12}$ contains 3. Then it contains $3 + 3 = 6$, then $6 + 3 = 9$, and the inverses 9 and 6. The set $\{0, 3, 6, 9\}$ is a subgroup, and it is the smallest subgroup that contains 3.

The same idea works in any group. If $a \in G$ and H is a subgroup containing a, then H must contain every power a^n with $n \in \mathbb{Z}$: the positive powers by closure, $a^0 = e$ as the identity, and the negative powers as inverses. The powers need not all be different. In V, $a^2 = e$, $a^3 = a$, $a^4 = e$, and $a^{-1} = a$.

Theorem 5.19. Let $a \in G$. Then $H = \{a^n \mid n \in \mathbb{Z}\}$ is a subgroup of G. It is the smallest subgroup of G that contains a: every subgroup containing a contains H.

Proof in brief. $a^r a^s = a^{r+s}$, so H is closed. $a^0 = e \in H$. And a^{-r} is in H with $a^{-r}a^r = e$. By Theorem 5.12, $H \leq G$. Any subgroup containing a contains all of H, so H is the smallest.

The footnote in the book makes a careful distinction. A subset is *minimal* with a property if no smaller subset has it. A subset is the *smallest* if it is contained in every subset with the property. $\{e, a\}$, $\{e, b\}$ and $\{e, c\}$ are all minimal nontrivial subgroups of V, but V has no smallest nontrivial subgroup.

DEFINITION 5.20 Let $a \in G$. The subgroup $\{a^n \mid n \in \mathbb{Z}\}$ of Theorem 5.19 is the **cyclic subgroup of G generated by a**, written $\langle a \rangle$.

Example 5.21. In D_{10}, $(\mu\rho^k)^2 = \iota$ and $(\mu\rho^k)^{-1} = \mu\rho^k$. So $\langle \mu\rho^k \rangle = \{\iota, \mu\rho^k\}$. Since $\rho^{10} = \iota$ and $\rho^{-1} = \rho^9$, we get $\langle \rho \rangle = \{\iota, \rho, \rho^2, \ldots, \rho^9\}$.

DEFINITION 5.22 An element **a generates G** and is a **generator for G** if $\langle a \rangle = G$. A group G is **cyclic** if some element of G generates it.

Example 5.23. $\mathbb{Z}_4$ is cyclic. Both 1 and 3 generate it: $\langle 1 \rangle = \langle 3 \rangle = \mathbb{Z}_4$. The group V is not cyclic, because each of $\langle a \rangle$, $\langle b \rangle$, $\langle c \rangle$ has only two elements.

Example 5.24. $\mathbb{Z}$ under addition is cyclic. Its only generators are 1 and -1. For $n \in \mathbb{Z}^+$, $\mathbb{Z}_n$ under addition modulo n is cyclic, and 1 and $n - 1$ are generators (there may be others).

Example 5.25. In $\mathbb{Z}$ under addition, $\langle 3 \rangle$ is the set of all multiples of 3. We write it $3\mathbb{Z}$. In the same way $n\mathbb{Z} = \langle n \rangle$. For example $6\mathbb{Z} < 3\mathbb{Z}$.

Example 5.26. U_n is the set of nth roots of unity. They are equally spaced on the unit circle (Figure 5.27). The point $\zeta = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ generates U_n: raising ζ to successive powers moves counterclockwise and lands on every root in turn. So $U_n = \langle \zeta \rangle$ is cyclic.

Figure 5.27 is the picture described in Example 5.26. The points $1, \zeta, \zeta^2, \zeta^3$ and so on sit around the circle, with i marked at one point. The figure is not redrawn here.

Notes on this section

This section covers book items 5.1 to 5.26 and Figures 5.10 and 5.27, with Tables 5.8 and 5.9. Examples 5.24

to 5.26 are the end of the cyclic-subgroup subsection; they are in this module because they belong to the same subsection. Exercises are not in these notes except the ones the text names; each is printed directly after the item that names it. The book's own exercise set for Section 5 (Exercises 1 to 65) is not reproduced; it is in the excerpt.

Section 6: Cyclic Groups

Printed pp. 61–67 (6.1–6.21; Figures 6.3, 6.11, 6.12, 6.19)

Before you start

- Cyclic subgroups $\langle a \rangle$ and generators (Section 5, Theorem 5.19 and Definitions 5.20 and 5.22).
- Integers: dividing one integer by another with a remainder. Every integer n has the form $mq + r$. This is the division algorithm that opens the section.
- Isomorphisms: a one-to-one, onto map that keeps the operation (Section 2).

1. Cyclic groups are abelian

In Section 5 a cyclic group was defined as a group with a generator. This section studies cyclic groups in detail. The first fact is that they are always abelian.

> **THEOREM 6.1** Every cyclic group is abelian.

Why. Let $G = \langle a \rangle$. Any two elements are a^r and a^s. Then $a^r a^s = a^{r+s} = a^{s+r} = a^s a^r$. So G is abelian. ◆

From here on we keep multiplicative notation, even though these groups are abelian.

The *division algorithm* is school arithmetic. Divide n by m and you get a quotient q and remainder r with $0 \le r < m$. Multiplying by m gives the form we need.

2. The division algorithm and the greatest common divisor

> **THEOREM 6.2 (DIVISION ALGORITHM FOR $\mathbb{Z}$)** If m is a positive integer and n is any integer, there are unique integers q and r with $n = mq + r$ and $0 \le r < m$.
>
> *Idea of the proof.* Mark the multiples of m on a number line and place n. Either n is a multiple qm (take $r = 0$), or n lies between two multiples of m; let qm be the first multiple to the left of n and let r be the distance from qm to n. Uniqueness: a multiple of m to the left of n at distance less than m is unique. ◆

Figure 6.3. The number-line picture for the proof. Two cases are drawn: $n \ge 0$ with $q \ge 0$, and $n < 0$ with $q < 0$. In each the remainder r is the distance from qm to n. Labels as printed; not redrawn.

In the notation of Theorem 6.2, q is the **quotient** and r is the nonnegative **remainder**.

Example 6.4. For 38 divided by 7:
$38 = 35 + 3 = 7(5) + 3$. So $q = 5$ and $r = 3$.

Example 6.5. For -38 divided by 7:
$-38 = -42 + 4 = 7(-6) + 4$. So $q = -6$ and $r = 4$.

We shall use the division algorithm to show that a subgroup of a cyclic group is cyclic. The natural generator to try is a suitable power of the generator a.

> **THEOREM 6.6** A subgroup of a cyclic group is cyclic.
>
> *Idea of the proof.* Let $G = \langle a \rangle$ and $H \le G$. If $H = \{e\}$, it is cyclic. Otherwise let m be the smallest positive integer with $a^m \in H$. Put $c = a^m$. For any $b = a^n \in H$, write $n = mq + r$ with $0 \le r < m$. Then $a^r = (a^m)^{-q} a^n \in H$. Minimality of m forces $r = 0$, so $b = c^q$. Hence $H = \langle c \rangle$. ◆

Examples 5.24 and 5.25 showed that $\mathbb{Z}$ is cyclic and that $n\mathbb{Z}$ is the cyclic subgroup generated by n. Theorem 6.6 says these are the only subgroups of $\mathbb{Z}$.

> **COROLLARY 6.7** The subgroups of $\mathbb{Z}$ under addition are precisely the groups $n\mathbb{Z}$, for $n \in \mathbb{Z}$.

This gives a clean definition of the greatest common divisor. The set $H = \{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$ (Exercise 54), so it is cyclic with a positive generator d.

> **EXERCISE 54, SECTION 6 (NAMED AFTER COROLLARY 6.7)** Let r and s be integers. Show that $\{nr + ms \mid n, m \in \mathbb{Z}\}$ is a subgroup of $\mathbb{Z}$.

> **DEFINITION 6.8** Let r be a positive integer and $s \ge 0$ an integer. The positive generator d of $H = \{nr + ms \mid n, m \in \mathbb{Z}\}$ is the **greatest common divisor** of r and s, written $d = \gcd(r, s)$.

Why is d the greatest common divisor? Since $r = 1r + 0s$ and $s = 0r + 1s$ lie in $H = d\mathbb{Z}$, the number d divides both r and s. And since $d \in H$, we can write $d = nr + ms$. Any common divisor of r and

s divides the right-hand side, so it divides d. So d is the largest common divisor.

The identity $d = nr + ms$ is called **Bézout's identity**.

Example 6.9. The positive divisors of 42 are $1, 2, 3, 6, 7, 14, 21, 42$; those of 72 are $1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72$. So $\gcd(42, 72) = 6$, and $6 = (3)(72) + (-5)(42)$. Two integers are **relatively prime** when their gcd is 1.

The next fact is needed for subgroups of cyclic groups. It is labelled (1) in the book:

If r and s are relatively prime and $r \mid sm$, then $r \mid m$. (1)

Proof of (1): write $1 = ar + bs$, multiply by m to get $m = arm + bsm$. Both terms on the right are divisible by r, since $r \mid sm$. So $r \mid m$.

3. The structure of cyclic groups

THEOREM 6.10 Let G be cyclic with generator a. If G is infinite, then $G \cong \langle \mathbb{Z}, + \rangle$. If G has finite order n, then $G \cong \langle \mathbb{Z}_n, +_n \rangle$.

Proof in brief. **Case I**: no positive power of a is e. Then every element is a^m for exactly one $m \in \mathbb{Z}$, and $\phi(a^i) = i$ is an isomorphism onto $\mathbb{Z}$, since $\phi(a^i a^j) = i + j$. **Case II**: $a^m = e$ for some $m > 0$. Let n be the least such m. Then $a^0, a, \ldots, a^{n-1}$ are distinct and exhaust G, and $\psi(a^i) = i$ is an isomorphism onto $\mathbb{Z}_n$, since $a^i a^j = a^k$ with $k = i +_n j$. ◆

Figures 6.11 and 6.12. Each shows the elements of a cyclic group of order n spaced evenly around a circle. In 6.11 the points are labelled $a^0 = e, a, a^2, a^3, \ldots, a^{n-1}$; in 6.12 the same points carry the exponents $0, 1, 2, 3, \ldots, n-1$. Labels as printed; not redrawn.

Example 6.13. Think of a^h as the point h units counterclockwise from e. To multiply a^h by a^k, start at a^h and move k more units. Write $h + k = nq + r$ with $0 \le r < n$. Going around the circle q full times brings you back to the start, so you finish at a^r. This is the isomorphism between $\langle a \rangle$ and $\mathbb{Z}_n$.

So the order of an element a of any group is the least positive n with $a^n = e$, as promised earlier.

Example 6.14. The order of a k-cycle $\sigma = (a_1, a_2, \ldots, a_k)$ is k. Applying σ moves each entry to the next. Each entry returns to itself only after k applications, and not sooner.

4. Subgroups of finite cyclic groups

THEOREM 6.15 Let G be cyclic with n elements, generated by a, and let $b = a^s$. Then b generates a cyclic subgroup H of G with n/d elements, where $d = \gcd(n, s)$. Also, $\langle a^s \rangle = \langle a^t \rangle$ if and only if $\gcd(s, n) = \gcd(t, n)$.

Proof in brief. H has as many elements as the least $m > 0$ with $b^m = e$, that is with $n \mid ms$. Write $d = un + vs$; then n/d and s/d are relatively prime. The least m with $\frac{m(s/d)}{n/d} \in \mathbb{Z}$ is n/d, by (1). Thus $|H| = n/d$. Since $\mathbb{Z}_n$ has only one subgroup of each order n/d, the second statement follows. ◆

Example 6.16. In $\mathbb{Z}_{12}$: $\gcd(3, 12) = 3$, so $\langle 3 \rangle = \{0, 3, 6, 9\}$ has $12/3 = 4$ elements. $\gcd(8, 12) = 4$, so $\langle 8 \rangle = \{0, 4, 8\}$ has 3 elements. $\gcd(5, 12) = 1$, so 5 generates all of $\mathbb{Z}_{12}$.

COROLLARY 6.17 If a generates a finite cyclic group G of order n, the other generators of G are the elements a^r with $\gcd(r, n) = 1$.

Example 6.18. Find the subgroups of $\mathbb{Z}_{18}$. By Corollary 6.17 the generators are $1, 5, 7, 11, 13, 17$. Next, $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10, 12, 14, 16\}$ has order 9, and its generators are $h2$ with h relatively prime to 9, namely $2, 4, 8, 10, 14, 16$. The element 6 of $\langle 2 \rangle$ generates $\{0, 6, 12\}$, and 12 also generates it. That accounts for $0, 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 16, 17$. The rest are $3, 9, 15$: $\langle 3 \rangle = \{0, 3, 6, 9, 12, 15\}$, which 15 also generates, and $\langle 9 \rangle = \{0, 9\}$.

Figure 6.19. The subgroup diagram of $\mathbb{Z}_{18}$. At the top is $\langle 1 \rangle = \mathbb{Z}_{18}$. Below it are $\langle 2 \rangle$ and $\langle 3 \rangle$. Below these are $\langle 6 \rangle$ and $\langle 9 \rangle$; $\langle 2 \rangle$ joins $\langle 6 \rangle$, $\langle 3 \rangle$ joins both $\langle 6 \rangle$ and $\langle 9 \rangle$. At the bottom is $\langle 0 \rangle$, joined to $\langle 6 \rangle$ and $\langle 9 \rangle$. Labels as printed; not redrawn.

The example is long because every step is shown. The exercises give practice with the same method.

COROLLARY 6.20 If G is a finite cyclic group and $H \le G$, then $|H|$ divides $|G|$.

Proof. H is cyclic by Theorem 6.6, generated by some $h = g^s$, where g generates G. By Theorem 6.15, $|H| = n/\gcd(n, s)$, which divides n. ◆

Example 6.21. The possible orders of subgroups of $\mathbb{Z}_{28}$ are the divisors of $28 = 2^2 \cdot 7$: $1, 2, 4, 7, 14, 28$. Each occurs: $\langle 0 \rangle$ has 1, $\langle 14 \rangle$ has 2, $\langle 7 \rangle$ has 4, $\langle 4 \rangle$ has 7, $\langle 2 \rangle$ has 14, and $\langle 1 \rangle = \mathbb{Z}_{28}$ has 28.

Corollary 6.20 holds for every finite group, not only cyclic ones. Lagrange's Theorem (Section 10 of the book, outside this module) gives that in general.

Notes on this section

This section covers book items 6.1 to 6.21 and Figures 6.3, 6.11, 6.12 and 6.19, with the relative-primeness statement labelled (1). Items 6.18 to 6.21 are included because they finish the subsection on subgroups of finite cyclic groups. The book's own exercise set (Exercises 1 to 67) is not reproduced here, apart from Exercise 54, which the text names; it is printed right after Corollary 6.7. Section 7 (Generating Sets and Cayley Digraphs) is Module V and is not in these notes.

Section 8: Groups of Permutations (8.1–8.18)

Printed pp. 77–82 (8.1–8.18; Table 8.9; Historical note)

Before you start

- Isomorphisms (Section 2): a one-to-one, onto map that keeps the operation.
- Functions: composition, one-to-one and onto maps, and inverse functions. Images and inverse images of sets (Definition 8.4 below).
- Permutations and cycles: the notation $(a_1, a_2, \ldots, a_k)$ for a cycle, and the group S_n of permutations of $\{1, \ldots, n\}$. The Section 4 material is used without reproduction.

1. Homomorphisms

An isomorphism keeps the operation and is one-to-one and onto. A **homomorphism** keeps the operation and need not be either. So every isomorphism is a homomorphism, but not the other way round.

> **DEFINITION 8.1** Let $\phi : G \to G'$ be a map between groups. ϕ is a **homomorphism** if $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G$.

Example 8.2. The map $\phi : \mathbb{R} \to U$ given by $\phi(x) = \cos(2\pi x) + i\sin(2\pi x) = e^{2\pi i x}$ is a homomorphism. The identity formulas for sines and cosines give $\phi(a+b) = \phi(a)\phi(b)$. It maps onto U but is not one-to-one, so it is not an isomorphism. Also $\phi(-x) = (\phi(x))^{-1}$.

Example 8.3. $\phi : U_{28} \to U_4$ given by $\phi(z) = z^7$ is well defined and a homomorphism: $(z_1 z_2)^7 = z_1^7 z_2^7$, and $z^{28} = 1$ gives $(z^7)^4 = 1$.

> **PLUS TWO RECALL: FUNCTIONS, IMAGES AND INVERSE IMAGES (STATUS OPEN, CHECK AGAINST THE KERALA TEXT)**
>
> **Recall.** For a map $\phi : X \to Y$, a subset A of X has an image $\phi[A]$ and a subset B of Y has an inverse image $\phi^{-1}[B]$. The inverse image can be nonempty even when ϕ is not onto.
>
> **Re-derive.**
>
> 1. Take $\phi(x) = x^2$ on $\mathbb{R}$, $B = \{4\}$. Then $\phi^{-1}[B] = \{2, -2\}$, although 4 is in the image.
> 2. Take $A = \{-1, 1\}$. Then $\phi[A] = \{1\}$.
>
> **Saved.** Definition 8.4.

> **DEFINITION 8.4** Let $\phi : X \to Y$, $A \subseteq X$, $B \subseteq Y$. The **image of A** is $\phi[A] = \{\phi(a) \mid a \in A\}$. The **inverse image of B** is $\phi^{-1}[B] = \{a \in X \mid \phi(a) \in B\}$.

The book gives proofs of the four properties in Theorem 8.5 because they are not obvious for maps that are neither one-to-one nor onto. For an isomorphism they are obvious: an isomorphism only relabels.

> **THEOREM 8.5** Let $\phi : G \to G'$ be a homomorphism.
>
> 1. $\phi(e) = e'$, the identity of G'.
> 2. $\phi(a^{-1}) = \phi(a)^{-1}$ for every $a \in G$.
> 3. If $H \leq G$, then $\phi[H] \leq G'$.
> 4. If $K' \leq G'$, then $\phi^{-1}[K'] \leq G$.
>
> *Proof in brief.* (1) $\phi(e) = \phi(ee) = \phi(e)\phi(e)$; cancel. (2) $e' = \phi(aa^{-1}) = \phi(a)\phi(a^{-1})$. (3) $\phi(a)\phi(b) = \phi(ab)$ keeps $\phi[H]$ closed; (1) and (2) give the identity and inverses. (4) If $a, b \in \phi^{-1}[K']$ then $\phi(ab) = \phi(a)\phi(b) \in K'$, and $\phi(a^{-1}) = \phi(a)^{-1} \in K'$; also $\phi(e) = e' \in K'$. ◆

With $\{e'\}$ in place of K', Theorem 8.5 (4) says $\phi^{-1}[\{e'\}]$ is a subgroup of G.

DEFINITION 8.6 For a homomorphism $\phi : G \to G'$ the subgroup $\phi^{-1}[\{e'\}] = \{x \in G \mid \phi(x) = e'\}$ is the **kernel** of ϕ, written $\mathbf{Ker}(\phi)$.

The kernel is used later in the book to define the alternating group, and the case $H = G$ of Theorem 8.5 (3) says $\phi[G]$ is a subgroup of G'. That case is used in Cayley's Theorem.

Example 8.7. In Example 8.2 the kernel of ϕ is the set of integers, since $e^{2\pi i x} = 1$ exactly when x is an integer. For $n \geq 1$, $\phi[\langle \frac{1}{n} \rangle] = U_n$.

Example 8.8. $\phi : \mathbb{Z}_n \to D_n$ given by $\phi(k) = \rho^k$ is a homomorphism. The two cases $a + b < n$ and $a + b \geq n$ are checked separately, using $\rho^n = \iota$. Its image is $\langle \rho \rangle$.

2. Cayley's Theorem

Many groups we have met are isomorphic to groups of permutations. $\mathbb{Z}_n$ is isomorphic to $\langle (1, 2, \ldots, n) \rangle \leq S_n$. D_n is a group of permutations of $\mathbb{Z}_n$. $GL(n, \mathbb{R})$ permutes the vectors of $\mathbb{R}^n$. A subgroup of a permutation group is called a **group of permutations**. Cayley's Theorem says every group is isomorphic to one.

The idea comes from the group table. Each row contains every element once. So each row defines a permutation of the group: put the table head on top and the row for a below. For D_3 (Table 8.9) the row for $\mu\rho$ gives

$$\begin{pmatrix} \iota & \rho & \rho^2 & \mu & \mu\rho & \mu\rho^2 \\ \mu\rho & \mu\rho^2 & \mu & \rho^2 & \iota & \rho \end{pmatrix}.$$

Table 8.9. The group table of D_3. Rows and columns are headed $\iota, \rho, \rho^2, \mu, \mu\rho, \mu\rho^2$ in that order:

D_3	ι	ρ	ρ^2	μ	$\mu\rho$	$\mu\rho^2$
ι	ι	ρ	ρ^2	μ	$\mu\rho$	$\mu\rho^2$
ρ	ρ	ρ^2	ι	$\mu\rho^2$	μ	$\mu\rho$
ρ^2	ρ^2	ι	ρ	$\mu\rho$	$\mu\rho^2$	μ
μ	μ	$\mu\rho$	$\mu\rho^2$	ι	ρ^2	ρ
$\mu\rho$	$\mu\rho$	$\mu\rho^2$	μ	ρ^2	ι	ρ
$\mu\rho^2$	$\mu\rho^2$	μ	$\mu\rho$	ρ	ρ^2	ι

Let λ_x be the permutation given by row x: $\lambda_x(g)$ is the entry in row x, column g, that is xg. This makes sense for infinite groups too.

DEFINITION 8.10 Let G be a group. The map $\phi : G \to S_G$ given by $\phi(x) = \lambda_x$, where $\lambda_x(g) = xg$ for all $g \in G$, is the **left regular representation** of G.

Each λ_x is a permutation. It is one-to-one, because $xa = xb$ gives $a = b$ by cancellation. It is onto, because $\lambda_x(x^{-1}b) = b$.

THEOREM 8.11 (CAYLEY'S THEOREM) Every group is isomorphic to a group of permutations.

Proof in brief. ϕ is one-to-one: if $\lambda_a = \lambda_b$ then $ae = be$, so $a = b$. ϕ is a homomorphism: $\lambda_{ab}(g) = (ab)g = a(bg) = (\lambda_a \lambda_b)(g)$. By Theorem 8.5 (3) with $H = G$, $\phi[G]$ is a subgroup of S_G, and $\phi : G \to \phi[G]$ is an isomorphism. ◆

Example 8.12. The proof gives a subgroup of S_G for any G, but not always the smallest one. D_n is isomorphic to a subgroup of $S_{\mathbb{Z}_n}$, although the proof gives a subgroup of S_{D_n} (which has $2n$ elements, not n). Also $\mathbb{Z}_6$ is isomorphic to a subgroup of S_5, since $(1, 2, 3)(4, 5)$ has order 6.

The **right regular representation** is defined next.

HISTORICAL NOTE Arthur Cayley (1821–1895) gave an abstract-sounding definition of a group in 1854, and in 1878 stated what is now Theorem 8.11 in a paper that argued from the group table. Walther von Dyck's axiomatic definition of 1882 gave the abstract theory its start. The book's full note is kept in the excerpt.

DEFINITION 8.13 Let G be a group. Put $\sigma_x(g) = gx$. The map $\tau : G \to S_G$ given by $\tau(x) = \sigma_{x^{-1}}$ is the **right regular representation** of G.

The right regular representation also proves Cayley's Theorem. The book leaves the details to Exercise 54, which is printed right after this paragraph.

EXERCISE 54, SECTION 8 (NAMED AFTER DEFINITION 8.13) Prove Cayley's Theorem using the right regular representation rather than the left regular representation.

3. Transpositions and orbits

DEFINITION 8.14 A cycle of length 2 is a **transposition**.

A transposition fixes every element except two, which it swaps. A computation shows

$$(a_1, a_2, \ldots, a_n) = (a_1, a_n)(a_1, a_{n-1}) \cdots (a_1, a_3)(a_1, a_2).$$

So every cycle of length n is a product of $n - 1$ transpositions.

> **THEOREM 8.15** Every permutation of a finite set with at least two elements is a product of transpositions.

In plain terms: any rearrangement of n objects can be produced by swapping pairs.

Example 8.16. $(1, 6)(2, 5, 3) = (1, 6)(2, 3)(2, 5)$.

Example 8.17. In S_n with $n \geq 2$, the identity is $(1, 2)(1, 2)$.

A product of transpositions for a given permutation is not unique (we can always insert $(1, 2)(1, 2)$). Later in the book it is shown that the number of transpositions is always even or always odd. That proof uses orbits, which are defined here.

For $\sigma \in S_A$ and $a \in A$, the **orbit** of a is $\{\sigma^k(a) \mid k \in \mathbb{Z}\}$. For a permutation in disjoint cycle form, the orbit of a is the set of entries in the cycle that contains a.

Example 8.18. Let $\sigma = (1, 2, 6)(3, 5) \in S_6$. The orbit of 1 is $\{1, 2, 6\}$, and the same set is the orbit of 2 and of 6. The orbit of 3 is $\{3, 5\}$. The orbit of 4 is $\{4\}$, since $\sigma = (1, 2, 6)(3, 5)(4)$ when 1-cycles are included.

Notes on this section

This section covers book items 8.1 to 8.18 and Table 8.9, with the historical note. Items 8.1 to 8.6 are homomorphisms and kernels (syllabus item 11); items 8.7 to 8.18 are Cayley's theorem, the regular representations, transpositions and orbits (syllabus item 12). The book's Section 8 continues past 8.18 into even and odd permutations, the alternating groups (8.19 to 8.25), which belong to Module III, and is not in these notes. The exercises of Section 8 are not reproduced except Exercise 54, which is printed after the text that names it.